

On the fly calibration and imaging of streamed TART data

Landman Bester

June 10, 2026

Abstract

This note develops the calibration and imaging machinery underpinning the proposed transient detector software pipeline; the detection heuristics themselves are derived in an accompanying document. Because those heuristics apply to calibrated, source-subtracted residual images, on-the-fly direction-independent calibration and calibrator subtraction are a necessary precursor. TART is a single-polarisation, zenith-pointing all-sky array, so we track one scalar complex gain per antenna, written as $g_p = \exp(a_p + i\psi_p)$, together with one real, non-negative flux per catalogued calibrator. The slowly varying log-amplitudes a_p , the faster varying phases ψ_p , and the source fluxes A_s are each modelled as independent integrated Wiener processes (IWPs); see the accompanying document for IWP notation and theory. The driving variances are *not* inferred online—they are calibrated once, offline, from the large body of existing data reductions—which avoids the ill-conditioned coupling between hyperparameter inference and a heavy-tailed likelihood. Conditioned on the gains the model visibility is *linear* in the fluxes, so the fluxes are marginalised out in closed form—a Rao–Blackwell-style elimination of the linear block, kept deliberately *unconstrained* so that the marginalisation is exact; physical non-negativity is imposed only when the subtraction model is formed (Section 5)—while the gains are updated by a robust iterated extended Kalman filter: an antenna-based Gauss–Newton step in (a, ψ) . The complex-gain StEFCal solve is retained, but only to *initialise* the filter, not as the per-frame iteration. The recovered flux scale is relative (self-calibrated): with no absolute flux reference in the catalogue, the gauge is fixed by *eliminating* a single reference antenna’s log-amplitude and phase from the state—a hard fix that, unlike an informative prior, cannot drift under accumulated process noise. Robustness to the inevitably incomplete satellite catalogue comes from a heavy-tailed (Student- t) observation model, realised as iteratively reweighted least squares. A short fixed-lag smoother refines the gain and flux estimates used for subtraction, after which the filter emits gain-corrected, model-subtracted residual visibilities, which are imaged onto a HEALPix grid to produce the residual image stream consumed by the detector.

1 Overview of the imaging and calibration problem

The task is to transform the raw uncalibrated visibilities $V_{pq,t}(\mathbf{b})$ into calibrated residual images $I_t(\mathbf{l})$ in real time, where p, q index the antennas, t the time, $\mathbf{b}$ the baseline vector, and $\mathbf{l}$ the direction on the sky. Because real-time streaming is required, calibration and calibrator source flux estimation needs to happen simultaneously (it is only the calibrator positions which are known a-priori and these are not always 100% accurate). The streaming nature of the problem makes Holoscan a natural fit for the software implementation, with CuPy—rather than an autodiff framework such as JAX—as the numerical backend for the inference machinery. Two considerations are decisive, and neither is that JAX *cannot* do the job—it offers DLPack zero-copy and its own memory-fraction controls. First, the calibration Jacobian is highly structured (Section 3): the antenna-bipartite gain block and the dense flux block are assembled into the normal equations by hand either way, so automatic differentiation buys little. Second, the pipeline must eventually run

on constrained hardware, where the real hazard is three GPU allocators—XLA’s, CuPy’s, and Holoscan’s—contending for the same device memory in one process; standardising on CuPy keeps a single, explicitly budgeted pool that already backs the existing Holoscan operators and exchanges buffers with them at zero copy. A third consideration is strategic: the TART calibration problem is, in absolute terms, tiny ($P \sim 24$ antennas, ~ 276 baselines, $S_t \sim 10\text{--}15$ visible sources) and would comfortably run on a CPU; but this pipeline is a *pathfinder* for the same architecture on much larger interferometric arrays, so every stage—including the calibration filter—is implemented GPU-resident from the outset, and structural choices (block layouts, structure-aware solvers) are made with scaling in mind rather than with TART’s trivial sizes as the design point. We nonetheless maintain a JAX reference implementation of the update for development and as a gradient-check oracle against the hand-coded CuPy kernels (Section 8).

We assume a measurement model of the form

$$V_{pq,t}(\mathbf{b}) = \exp(a_p(t) + \psi_p(t)) \left(\sum_s A_s(t) X_{pq,ts} \right) \exp(a_q(t) - \psi_q(t)), \quad (1)$$

where the scalar gain log-amplitudes $a_{p/q}$ are slowly varying real functions of time, the gain phases $\psi_{p/q}$ are more quickly varying real functions of time, the calibrator source fluxes A_s are slowly varying real functions of time, and $X_{pq,ts}$ is the known unit coherencies of source s to the visibility on baseline pq at time t . The TART telescope [7] always points at zenith and sees almost the whole hemisphere it is located in. This makes spherical coordinates the natural choice for the image domain and the calibrator source positions. The residual images should be computed on a Healpix grid.

Calibrator sources are predominantly Global Navigation Satellite System (GNSS) satellites in medium Earth orbit (MEO, $\sim 20,200$ km), with a minority in geostationary or inclined-geosynchronous orbits along the ecliptic. They are extremely bright (of order $10^5\text{--}10^7$ Jy) and have well-known positions from broadcast ephemerides, though their fluxes vary on minute-to-hour time-scales, and their orbital motion (negligible only for the geostationary minority) is what modulates each fringe and aids the flux/gain separation. These sources dominate the L1 sky by orders of magnitude over natural emission: the Sun is comparatively faint at 1.575 GHz and contributes appreciably only during active periods, and the A-team sources sit below this array’s confusion floor. The catalogue is therefore satellite-driven, but the source machinery is agnostic to source type—folding in the Sun (known ephemeris, non-negative IWP flux) is a sensible first extension, with bright catalogued sources a later milestone—and any extended Galactic emission, which the point-source model cannot represent, will appear in the residual as a sidereally drifting background (a consideration for the detector’s choice of pixel frame). The gain log-amplitudes and phases are driven by independent integrated Wiener processes (IWPs) with different driving noise levels. The calibrator source fluxes are also driven by independent IWPs, with smaller driving noise levels than the gain parameters—though “smaller” must be qualified: what the filter tracks is the *apparent* flux, which for a moving satellite is modulated by the antenna element beam and the satellite’s own transmit pattern on the same minute time-scales as its elevation change (see Section 6 for the implied constraint on σ_A^2 and the planned beam-model mitigation). Since the catalogue is almost always incomplete, a robust heavy-tailed likelihood is required to prevent un-modelled *point-like* outliers (a passing un-catalogued emitter, RFI) from dominating inference—note that this mechanism handles sparse per-baseline outliers, not a bright coherent source, which must instead be added to the catalogue. The IWP state-space model and the heavy-tailed likelihood are combined in a robust iterated extended Kalman filter: the gains are updated by an antenna-based Gauss–Newton step in (a, ψ) , while the catalogue fluxes—which enter the model linearly—are marginalised out in closed

form. The filter tracks the calibration parameters, subtracts the calibrator sources, and emits a gain-corrected residual visibility stream that is imaged to produce the residual images fed to the detection pipeline.

The remainder of this note formalises the calibration filter and the imaging step. Section 2 sets up the IWP state-space model and fixes the gauge; Section 3 gives the linearised (iterated-EKF) observation update and its Jacobian; Section 4 adds the heavy-tailed likelihood; Section 5 Rao–Blackwellises the source fluxes and describes their warm-up; Section 6 defines the gain-corrected residual visibilities and their projection onto a HEALPix grid; Section 7 covers offline calibration of the driving variances; and Section 8 collects implementation choices and reuse of existing TART software (the `tart-telescope` organisation and T. Molteni’s repositories, especially `DISKO` and `SPOTLESS`).

2 State-space model for the calibration parameters

Let P be the number of antennas and S_t the number of catalogued sources above the horizon at time t . We track the gain log-amplitudes a_p , the gain phases ψ_p ($p = 1, \dots, P$), and the source fluxes A_s ($s = 1, \dots, S_t$). Each scalar parameter is an independent IWP [3]; following the accompanying note we lift a scalar parameter θ and its first q derivatives into a state $\boldsymbol{\theta} = (\theta, \dot{\theta}, \dots, \theta^{(q)})^\top$ with the exact discrete transition

$$\boldsymbol{\theta}_k = \mathbf{A}_\theta(\Delta_k) \boldsymbol{\theta}_{k-1} + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_\theta(\Delta_k)), \quad (2)$$

where $\mathbf{A}_\theta, \mathbf{Q}_\theta$ are the closed-form IWP matrices of the accompanying document and depend on the per-parameter driving variance σ_θ^2 . We default to $q = 1$ (a locally linear, “constant-velocity” model) for every parameter; the order may be raised per group if the offline data warrant it. Stacking the per-parameter blocks gives the full state $\mathbf{x}_k$ and a block-diagonal transition

$$\mathbf{x}_k = \mathbf{F}_k \mathbf{x}_{k-1} + \mathbf{w}_k, \quad \mathbf{F}_k = \text{blkdiag}(\{\mathbf{A}_{a_p}\}, \{\mathbf{A}_{\psi_p}\}, \{\mathbf{A}_{A_s}\}), \quad (3)$$

with $\mathbf{Q}_k$ the matching block-diagonal process covariance, so the predict step is exact, decoupled across parameters, and $O(P + S_t)$ per frame.

Irregular cadence is a hard requirement. The transition matrices are functions of the *actual* inter-frame interval Δ_k , and this is precisely what makes the filter robust to dropped frames, ingest hiccups, and variable cadence: gaps are handled exactly, for free, by evaluating $\mathbf{A}_\theta(\Delta_k)$ and $\mathbf{Q}_\theta(\Delta_k)$ from the timestamp stream every frame. The implementation must therefore, under no circumstances, assume a constant Δ —no stage may cache fixed-step transition or process-noise matrices, and every operator consumes the timestamps alongside the data. Any fixed-step shortcut silently forfeits this robustness and turns every dropped frame into a model-mismatch transient.

Driving variances are fixed. The driving-variance groups— σ_a^2 (slow), σ_ψ^2 (faster), and σ_A^2 (slow, modulo the warm-up of Section 5)—are *not* estimated online. They are calibrated once from the existing offline reductions (Section 7). This is deliberate: inferring the driving variances through the marginal likelihood while a heavy-tailed likelihood simultaneously reweights the data is poorly conditioned, the two mechanisms competing to explain the same residual.

Identifiability and gauge fixing. The likelihood depends on the gains only through the products $g_p g_q^* = \exp(a_p + a_q) \exp(i(\psi_p - \psi_q))$, which leaves exactly two unobservable real degrees of freedom (rank-one each), independent of source motion:

- a global phase $\psi_p \mapsto \psi_p + \phi$ cancels in every $g_p g_q^*$; we remove it by *eliminating* a reference-antenna phase from the state, $\psi_{\text{ref}} \equiv 0$, so the freedom is absent in both mean and covariance;
- a global log-amplitude $a_p \mapsto a_p + c$ rescales every visibility by e^{2c} and is exactly degenerate with an overall flux rescaling $A_s \mapsto e^{-2c} A_s$. The catalogue supplies positions but *no* fluxes, so the scale cannot be anchored on the source side. We fix it exactly as we fix the phase: by *eliminating* the reference antenna’s log-amplitude from the state, $a_{\text{ref}} \equiv 0$ (i.e. unit reference amplitude), using the *same* reference antenna for both gauges. Two rejected alternatives explain the choice. The arbitrary zero-sum constraint $\sum_p a_p = 0$ spreads a fictitious constraint across all antennas and, enforced as a mean projection, leaks process noise into the unconstrained covariance direction. An informative Gaussian prior on each a_p (centred on offline-determined means) is more subtly defective: in a *filter*, a prior is only the initial condition. The all-ones amplitude mode is unobservable, so it is never corrected by data, and every predict step injects process noise into it—the covariance along that mode grows without bound and the mean random-walks. The prior anchors the gauge at $t = 0$ and nothing thereafter; restoring persistence would require re-applying it as a per-frame pseudo-observation, which is workable but adds machinery. Eliminating a_{ref} breaks exactly the rank-one scale freedom, leaves the relative antenna amplitudes entirely data-driven, cannot drift, and is the precise analogue of the reference-phase elimination—one mechanism, applied twice. The offline amplitude means are retained, but demoted to what they really are: a good *initialisation* (re-centred so that $a_{\text{ref}} = 0$; Section 3), not a gauge.

The recovered flux scale is therefore *relative* (self-calibrated), pinned to the reference antenna’s true (unmodelled) gain amplitude; this is sufficient for transient detection, which is differential. The per-frame noise scale entering the detector is independent of this gauge (Section 6).

Reference-antenna failover. Tying both gauges to one antenna creates an operational single point of failure: if the reference antenna is flagged or fails mid-run, both gauges are lost. In principle the state can be re-gauged analytically—subtract the new reference’s phase and log-amplitude from all antennas’ means and apply the corresponding congruent transform to the covariance—and this is the eventual refinement. For the initial implementation we adopt the simpler, more conservative procedure: *reset* the pipeline with a new reference antenna, re-running the StEFCal acquisition of Section 3. Acquisition is fast (a few bootstrap iterations) and reference failure should be rare, so the occasional re-acquisition transient is an acceptable price for not maintaining re-gauging code. The choice of reference antenna, and the failover order, are configuration items; the monitoring stream (Section 8) should alert on reference-antenna flagging specifically.

3 Observation model and its linearisation

At time t_k each baseline pq contributes a complex visibility. With the single-polarisation scalar gains and the catalogue coherencies $X_{pq,s}$ —unit-amplitude fringes computed from the known source directions (Section 6)—the noise-free model is

$$\hat{V}_{pq} = g_p \hat{M}_{pq} g_q^*, \quad \hat{M}_{pq} = \sum_s A_s X_{pq,s}, \quad g_p = \exp(a_p + i\psi_p). \quad (4)$$

We observe $V_{pq} = \hat{V}_{pq} + n_{pq}$ with circular complex noise n_{pq} whose variance is set by the data weights, stacking real and imaginary parts as two real observations per baseline.

StEFCal for acquisition, (a, ψ) for tracking. StEFCal [4] exploits that, holding all *other* antennas’ complex gains fixed, $\hat{V}_{pq}$ is *linear* in g_p , giving an antenna-by-antenna linear least squares. Its real virtue is that this complex-linear form has no phase-wrapping or local-minimum problem, which makes it an excellent *acquisition* (cold-start) solver. We do not, however, run it as the per-frame iteration. We track the gains as (a_p, ψ_p) for two reasons that the complex- g parametrisation does not serve: the log-amplitude and phase carry distinct temporal statistics (slow amplitude, faster phase) and so want *separate* IWP driving variances; and the log enforces amplitude positivity intrinsically. $\hat{V}_{pq}$ is mildly nonlinear in (a, ψ) , so we linearise directly (Gauss–Newton / iterated EKF). The antenna-based normal-equations structure StEFCal exploits survives the change of parametrisation (Section 3), in a cleaner form; the cost is that the phase linearisation is valid only while the frame-to-frame phase change stays well under $\pm\pi$, which the frame rate must respect.

Initialisation. The IWP-EKF needs a good initial state, and a cold-start phase solution is non-convex. We bootstrap it with StEFCal, exploiting that the StEFCal *phases* are driven by the known source geometry and are reliable even with a rough flux model, whereas the StEFCal *amplitudes* are entangled with the unknown fluxes and are not. The recipe is: (i) seed a rough catalogue flux model; (ii) solve complex gains with StEFCal and keep only the phases (re-referenced so $\psi_{\text{ref}} = 0$); (iii) set the antenna log-amplitudes to their offline-determined mean values *re-centred on the reference antenna*, $a_p \leftarrow \bar{a}_p - \bar{a}_{\text{ref}}$, so that $a_{\text{ref}} = 0$ holds by construction—the offline means serve only as a good starting point, the gauge itself being fixed by the elimination of Section 2; (iv) with the gains thus fixed, solve the linear least-squares problem for the source fluxes. Steps (ii)–(iv) may be iterated two or three times—this is simply bootstrapped self-calibration and converges quickly—which makes the phases robust to the initial relative-flux guess. The resulting gains and fluxes seed the IWP-EKF mean, with the offline-calibrated covariances as the initial uncertainty.

Jacobian. Differentiating (4) at the current estimate,

$$\frac{\partial \hat{V}_{pq}}{\partial a_p} = \frac{\partial \hat{V}_{pq}}{\partial a_q} = \hat{V}_{pq}, \quad \frac{\partial \hat{V}_{pq}}{\partial \psi_p} = +i \hat{V}_{pq}, \quad \frac{\partial \hat{V}_{pq}}{\partial \psi_q} = -i \hat{V}_{pq}, \quad \frac{\partial \hat{V}_{pq}}{\partial A_s} = g_p X_{pq,s} g_q^*. \quad (5)$$

The gain derivatives are simply the model visibility rotated by 0 or $\frac{\pi}{2}$, so the Jacobian costs nothing beyond $\hat{V}_{pq}$, which we already form. Each baseline row touches only its two antennas: the gain block of the Jacobian inherits the *bipartite* (baseline $\times$ antenna) sparsity of the array’s incidence structure, whereas the flux block $\partial \hat{V}_{pq} / \partial A_s$ is dense (every baseline sees every source). The antenna-based normal-equations structure that StEFCal exploits therefore re-appears in $\mathbf{H}_k^* \mathbf{H}_k$, despite the change of parametrisation—which is what makes a hand-coded, structure-aware solver attractive.

Iterated-EKF measurement update. Let $\mathbf{H}_k$ be the stacked (real/imag) Jacobian (5) and $\mathbf{e}_k = V_k - \hat{V}_k(\mathbf{x}_{k|k-1})$ the innovation at the predicted mean. One Gauss–Newton step is the standard EKF update [2]; relinearising about the running estimate and repeating yields the *iterated* EKF, i.e. a Gauss–Newton solver for the MAP state under the IWP prior. The phase block in particular can converge slowly, so the inner iteration count is a tunable budget; in the spirit of starting simple we cap it, warm-start each frame from the previous solution (frame-to-frame

change is small for a tracking filter), and raise the cap only if the gain-block NIS indicates under-convergence. The normalised innovation (NIS) formed here is logged as the whiteness diagnostic [1] of the accompanying note, but—per Section 2—is never fed back to adapt the driving variances.

4 Robust update via a heavy-tailed likelihood

The Gaussian observation model of Section 3 is fragile: a baseline contaminated by an un-catalogued source or by radio-frequency interference (RFI) is a gross outlier that, under a Gaussian likelihood, would bias every gain and flux. We replace the Gaussian by a Student- t [5] observation density, which has the Gaussian scale-mixture representation

$$t_\nu(\mathbf{e} | \mathbf{0}, \mathbf{R}) = \int_0^\infty \mathcal{N}(\mathbf{e} | \mathbf{0}, \mathbf{R}/\lambda) \text{Gamma}(\lambda | \frac{\nu}{2}, \frac{\nu}{2}) d\lambda, \quad (6)$$

in which the latent per-baseline precision scale λ_{pq} down-weights discrepant data. Treating $\{\lambda_{pq}\}$ by expectation–maximisation (equivalently, one variational step) gives an iteratively reweighted least squares (IRLS) update: at Gauss–Newton iteration j each baseline carries a weight

$$w_{pq}^{(j)} = \frac{\nu + n_y}{\nu + \epsilon_{pq}^{(j)}}, \quad \epsilon_{pq}^{(j)} = \mathbf{e}_{pq}^{(j)\top} \mathbf{R}_{pq}^{-1} \mathbf{e}_{pq}^{(j)}, \quad (7)$$

with $n_y = 2$ for the stacked real/imag residual, and the EKF normal equations are assembled with the reweighted observation precision $w_{pq} \mathbf{R}_{pq}^{-1}$. Inliers keep $w_{pq} \approx 1$; outliers are smoothly suppressed. The degrees-of-freedom ν is fixed (a small value, $\nu \sim 4$, is robust) rather than inferred, again to avoid the coupling of Section 2. Two caveats. First, with only $n_y = 2$ degrees of freedom per baseline, ϵ_{pq} is a noisy (χ_2^2 -like) statistic and the weights themselves chatter from frame to frame even on clean data; this is standard and largely harmless for the gain solution, but it argues against pushing ν much below 4, where the weight function steepens and the chatter amplifies. Second, and critically: the IRLS weights are *internal to the calibration update* and must, under no circumstances, propagate into the imaging weights of Section 6. A genuine transient is an un-catalogued coherent source—exactly what the Student- t machinery down-weights to protect the gains. That protection is correct for calibration, but if the same weights were applied when imaging the residuals, the pipeline would partially subtract its own science signal. The imaging weights are the (gain-corrected) thermal weights only.

Remark 1 (Two failure modes of an incomplete catalogue). *Robustness here and flux handling in Section 5 address different symptoms of catalogue incompleteness, and both are needed:*

- **Present but un-catalogued** sources have no position, hence no $X_{pq,s}$, and cannot be modelled; they appear as heavy-tailed outliers and are handled only by the Student- t down-weighting. Note that a bright transient—the very thing the downstream detector exists to find—falls in this class by design: the down-weighting protects the gain solution during the event, while the signal itself flows through unattenuated in the residual stream (the imaging-weight firewall above) and additionally announces itself as a coherent multi-baseline NIS excess (Section 8).
- **Catalogued but absent** sources do have an $X_{pq,s}$ but no real emission; they are handled by driving their flux state to zero (Section 5).

5 Rao–Blackwellised flux update and warm-up

Conditioned on the gains, the model (4) is *linear* in the fluxes: with g_p, g_q fixed, $\hat{V}_{pq} = \sum_s A_s (g_p X_{pq,s} g_q^*)$. Partition the state into the nonlinear gains $\mathbf{x}_g = \{a_p, \psi_p\}$ and the linear fluxes $\mathbf{x}_A = \{A_s\}$. The joint predicted density is Gaussian (block-diagonal, the IWPs being independent) and the flux-conditional posterior $p(\mathbf{x}_A | \mathbf{x}_g, V_k)$ is exactly Gaussian—we keep the flux block *unconstrained* precisely to preserve this exactness (see below). We therefore eliminate $\mathbf{x}_A$ analytically: rather than sampling or jointly Newton-stepping them, we marginalise them in closed form via the Schur complement of the (reweighted) normal-equations matrix, so the gain update sees the fluxes integrated out with their full uncertainty, and the flux posterior is recovered by back-substitution. This is an exact Gaussian marginalisation of the linear block (a Rao–Blackwell-*style* variance reduction achieved by conditioning, but without Monte Carlo—there is a single point estimate of the gains, not a particle ensemble, so the term is used only by analogy). It lowers the variance of the gain estimate and confines the per-frame nonlinear solve to the $2P$ -dimensional gain block. Because the flux block is an unconstrained symmetric positive-definite system, its solve (and the back-substitution) can use a plain conjugate-gradient iteration warm-started from the previous frame’s fluxes—no active-set or constrained-QP machinery anywhere in the per-frame path, which is also what keeps the update a fixed sequence of dense linear-algebra primitives amenable to a GPU implementation.

Non-negativity. A physical flux satisfies $A_s \geq 0$, but we deliberately do *not* impose this as a constraint on the filter state. The primary strategy is: the flux block stays fully Gaussian and unconstrained—so the Schur marginalisation above remains exact and the filter recursion never branches—and non-negativity is applied only at the point of *use*, by clipping when the subtraction model is formed,

$$\hat{M}_{pq} = \sum_s \max(\hat{A}_s, 0) X_{pq,s}. \quad (8)$$

A genuinely absent source then behaves as follows: with a zero-mean initialisation, the small post-warm-up σ_A^2 , and no likelihood gradient pulling it up, its flux estimate hovers in a small noise-level band around zero; the clip simply prevents the negative excursions of that band from being *added* to the data through the subtraction. Nothing the detector consumes is affected, the state remains exactly Gaussian, and there is no active-set bookkeeping to maintain or to break. (Earlier drafts imposed the constraint on the state itself via non-negative least squares with a moment-matched truncated-Gaussian posterior. That machinery is statistically tidier but operationally the most fragile component of the design—stateful branching as sources rise, set, and cross the boundary—and it destroys the exactness of the marginalisation. It is retained only as a fallback, to be revisited if offline replay shows material leakage that clipping does not control.) Should residual leakage into absent sources be observed, the first escalation is not the hard constraint but a mild ℓ_1 (LASSO-style proximal) shrinkage on $\mathbf{x}_A$, which supplies *active* suppression while keeping the update simple; we start without it and add it only if needed.

Warm-up schedule. At acquisition the fluxes are unknown and the gains are not yet locked, so we initialise every flux at mean zero with a diffuse covariance and run the flux driving variance σ_A^2 *large*, letting the data raise the fluxes of sources that are truly present. As the filter converges—monitored via the settling of the gain-block NIS rather than a fixed frame count— σ_A^2 is annealed down to the realistic floor calibrated offline (Section 7); an exponential schedule is the fallback. Because a large warm-up σ_A^2 also lets noise excite *absent* sources (which then freeze when the variance tightens), the zero-mean initialisation, the small post-warm-up driving variance, and the

Student- t weights together bias those fluxes back down, with the subtraction-time clip and the optional ℓ_1 escalation above as the backstops.

6 Calibrated residuals and all-sky imaging

Coordinate frame. TART points at zenith and does not track, so there is no moving phase centre and no fringe stopping. The natural, least error-prone choice is to work in a single fixed local east–north–up (ENU) frame and use the raw geometric phase $\mathbf{b} \cdot \hat{\mathbf{s}}$ throughout, with no $(n-1)$ reference term (which is merely a constant per-baseline zenith phase, harmless if applied consistently but pointless here). Because the catalogue returns source directions in azimuth/elevation, and HEALPix is coordinate-agnostic, we define the HEALPix grid directly in this frame: identify the HEALPix native Cartesian axes with ENU ($z = \text{up} = \text{zenith}$), so that `pix2vec` returns the pixel direction cosines $\hat{\mathbf{s}}_j = (l_j, m_j, n_j)$ with no tangent-plane projection, the horizon mask is simply $n_j = z_j \geq \sin(\text{el}_{\min})$, and a catalogue az/el converts to the same frame in one line. Three conventions are pinned once (azimuth zero/handedness, the ENU axis assignment, and that $\mathbf{b}_{pq} = (u_{pq}, v_{pq}, w_{pq})$ lives in this frame); everything downstream is then consistent.

Coherencies from the catalogue. The satellite catalogue gives positions (from broadcast ephemerides / two-line elements), not fluxes. For source s with time-dependent direction cosines $\hat{\mathbf{s}}_s(t) = (l_s, m_s, n_s)$ and baseline $\mathbf{b}_{pq} = (u_{pq}, v_{pq}, w_{pq})$, the unit coherency is the geometric fringe

$$X_{pq,s}(t) = \exp\left(2\pi i \frac{\nu}{c} \mathbf{b}_{pq} \cdot \hat{\mathbf{s}}_s\right) = \exp\left(2\pi i \frac{\nu}{c} [u_{pq}l_s + v_{pq}m_s + w_{pq}n_s]\right), \quad n_s = \sqrt{1 - l_s^2 - m_s^2}. \quad (9)$$

Satellites in medium Earth orbit move appreciably over minutes, so $X_{pq,s}$ is recomputed every frame from the propagated positions; the motion is in fact helpful, modulating each source’s fringe and so aiding the flux/gain separation (the geostationary minority move little and are correspondingly harder to separate from the gains). The motion has, however, a second consequence that the flux model must own: the IWP tracks the *apparent* flux $A_s(t)$, which is the intrinsic emission multiplied by the antenna element gain pattern and by the satellite’s own transmit pattern, both of which change with the source’s elevation—i.e. on the same minute time-scales as the orbit. This is a fast, largely *deterministic* trend, not slow stochastic drift, and a σ_A^2 frozen at a value fitted to “intrinsic” variability will lag it systematically; lag error multiplied by a 10^5 – 10^7 Jy source is precisely the residual-sidlobe contamination Section 8 warns about. The clean mitigation is to factor the deterministic part out, $A_s(t) = E(\text{el}_s(t)) \tilde{A}_s(t)$, with $E(\text{el})$ a static (azimuthally averaged) element-beam model fitted once offline—a per-source scalar multiplying $X_{pq,s}$, essentially free at runtime—so the IWP tracks only the genuinely stochastic factor $\tilde{A}_s$. An approximate TART element-beam model is the subject of an ongoing student project and this factorisation is therefore *left as future work*; until it lands, the offline fit of σ_A^2 (Section 7) must be performed on apparent-flux tracks so that the floor is set high enough to follow the beam-induced trend, accepting the attendant loss of flux/gain separability, and the bright-source masking of Section 8 carries the residual risk. The coordinate pipeline (ephemeris propagation, ECEF→topocentric az/el→direction cosines, horizon masking) is the natural place to reuse TART community code (Section 8); note that two-line-element accuracy ($\sim$ km at MEO, i.e. tens of arcseconds) is utterly negligible against TART’s degrees-wide point-spread function, so precise post-processed ephemerides are not required.

Fixed-lag smoothing for subtraction. The filtered estimate $\hat{\mathbf{x}}_{k|k}$ lags the true state by construction, and for the slowly varying amplitudes that lag is the dominant subtraction error once

the gauge and beam issues above are handled. Smoothing repairs it cheaply: a fixed-lag Rauch–Tung–Striebel (RTS) pass over a short window turns the causal estimate into $\hat{\mathbf{x}}_{k-L|k}$, conditioned on L frames of future data. Concretely, the calibration operator maintains a ring buffer of the last $L+1$ frames’ predicted and filtered moments $\{\hat{\mathbf{x}}_{j|j-1}, \mathbf{P}_{j|j-1}, \hat{\mathbf{x}}_{j|j}, \mathbf{P}_{j|j}, \mathbf{F}_j\}$ (the $\mathbf{F}_j$ stored per frame because Δ_j varies; Section 2); on receipt of frame k it runs the standard backward recursion

$$\mathbf{G}_j = \mathbf{P}_{j|j} \mathbf{F}_{j+1}^\top \mathbf{P}_{j+1|j}^{-1}, \quad \hat{\mathbf{x}}_{j|k} = \hat{\mathbf{x}}_{j|j} + \mathbf{G}_j (\hat{\mathbf{x}}_{j+1|k} - \hat{\mathbf{x}}_{j+1|j}), \quad (10)$$

from $j = k - 1$ down to $j = k - L$, and emits the residual for frame $k - L$ using the *smoothed* gains and (clipped) fluxes in (11). The filtered quantities retain their roles—warm-starting the next frame’s iterated update and feeding the NIS diagnostics, which are innovations-based and therefore inherently causal. The costs are explicit: $O(L)$ extra state memory, an $O(L n^3)$ backward pass per frame ($n = \dim \mathbf{x}$; trivial at TART scale, and for larger arrays $\mathbf{F}$ is block-diagonal so the only dense work is in the $\mathbf{P}$ products), and—the one that matters—a latency of L frames added to the residual stream and hence to detection. At TART’s seconds-level cadence a default of $L = 1\text{--}2$ is a negligible price; L is a configuration item, with $L = 0$ recovering the pure filter for latency-critical deployments.

Gain-corrected residual visibilities. After the smoothing pass the filter forms the model $\hat{M}_{pq} = \sum_s \max(\hat{A}_s, 0) X_{pq,s}$ from the smoothed estimates and emits, per baseline, the gain-corrected, model-subtracted residual together with its corrected weight,

$$R_{pq} = \hat{g}_p^{-1} V_{pq} \hat{g}_q^{-*} - \hat{M}_{pq}, \quad w_{pq}^{\text{corr}} = w_{pq} |\hat{g}_p \hat{g}_q|^2. \quad (11)$$

Here w_{pq} is the *thermal* weight—never the IRLS weight of Section 4, which is internal to calibration (a transient is exactly what IRLS down-weights, and propagating those weights into the imaging would partially subtract the science signal). TART data carry no per-visibility thermal weights, and the antennas are very nearly identical, so we assume i.i.d. noise: a single constant $w_{pq} = 1/\sigma_n^2$, with σ_n^2 estimated once, offline, from a large batch of archived data alongside the driving variances (Section 7). Working in the gain-corrected (sky) frame puts the residual on a consistent *relative* (self-calibrated) flux scale with a well-characterised noise variance—the basis for the per-frame variance R_k the detector consumes (made precise below). By construction R_{pq} has the same layout as a raw visibility, so the downstream imager need not know that calibration occurred.

HEALPix imaging (default: dirty map). The default residual map is the adjoint (gridless) DFT of R_{pq} evaluated at the visible HEALPix pixel centres [6]. On an *equal-area* grid the solid-angle element is constant, so there is *no* $1/n$ Jacobian (that factor belongs only to the $dl dm$ tangent-plane element, and on this grid it would merely re-inject a horizon singularity as $n_j \rightarrow 0$); and with no fringe stopping the phase is the raw $\mathbf{b} \cdot \hat{\mathbf{s}}_j$, with no $(n_j - 1)$:

$$I_t(\hat{\mathbf{s}}_j) = \text{Re} \sum_{pq} w_{pq}^{\text{corr}} R_{pq} \exp\left(-2\pi i \frac{\nu}{c} \mathbf{b}_{pq} \cdot \hat{\mathbf{s}}_j\right). \quad (12)$$

This is the natural representation for a zenith-pointing instrument that sees most of the hemisphere (a single tangent plane breaks down at large zenith angle). The existing tiled (l, m) DFT operator is a structural template for the per-pixel phase evaluation and memory tiling, but the radiometric normalisation is re-derived for the equal-area grid rather than ported—in particular the $1/n$ and $(n - 1)$ factors are dropped. Each frame yields one all-sky residual map; the stream of maps is the per-pixel light-curve input to the detector. The dirty map’s noise is correlated between pixels through the point-spread function, and $\{w_{pq}^{\text{corr}}\}$ gives only the per-pixel variance (the diagonal); the inter-pixel correlation is what the optional mode below addresses.

HEALPix imaging (optional: Tikhonov map). As a higher-fidelity, more resource-hungry alternative, the residual can be reconstructed by Tikhonov-regularised least squares rather than the bare adjoint,

$$\hat{\mathbf{I}}_t = (\mathbf{A}^* \mathbf{W} \mathbf{A} + \mu \mathbf{\Gamma})^{-1} \mathbf{A}^* \mathbf{W} \mathbf{R}, \quad (13)$$

with $\mathbf{A}$ the (constant, in the array frame) HEALPix DFT operator, $\mathbf{W} = \text{diag}\{w^{\text{corr}}\}$, and $\mu \mathbf{\Gamma}$ the regulariser ($\mathbf{\Gamma} = \mathbf{I}$, or a smoothing operator). This is the DiSkO [8] reconstruction and buys three things the dirty map lacks: a sampling-density-corrected point-source response (so a fixed detection threshold means the same thing across the sky), a partially decorrelated noise field, and—most usefully for the detector—the posterior covariance $(\mathbf{A}^* \mathbf{W} \mathbf{A} + \mu \mathbf{\Gamma})^{-1}$, whose diagonal furnishes a principled per-pixel R_k . It does not *fully* whiten (the Gram inverse is not diagonal for a sparse array), so the look-elsewhere caveats of the detector note are reduced, not removed. Because this per-pixel R_k scales with the regulariser, μ (and the choice of $\mathbf{\Gamma}$) must itself be calibrated offline and *frozen*, just as the driving variances are (Section 7): a per-frame retuning of μ would make the detector’s per-pixel noise scale—and hence its false-alarm rate—drift between frames. Cost is one linear solve per frame, but cheap for TART: in the array (zenith) frame $\mathbf{A}$ is constant, so $\mathbf{A}^* \mathbf{W} \mathbf{A}$ changes only through the slowly varying gain weights, and the solve is by conjugate gradients warm-started from the previous frame’s map (which barely changes for a mostly-noise residual). A pre-factorisation of $\mathbf{A}^* \mathbf{W} \mathbf{A}$ is deliberately *not* used: $\mathbf{W}$ drifts with the gains, and a stale factor silently biases both the map and—worse—the per-pixel R_k the detector’s false-alarm calibration rests on. The solved map may optionally be rotated onto an equatorial HEALPix grid for the detector, decoupling the constant-operator solve from the fixed-sky detection grid. We default to the dirty map and enable the Tikhonov mode where resources allow; running both against DiSkO is also our imaging validation.

7 Offline calibration of the driving variances

The driving variances are estimated once, offline, from the existing body of data reductions and then frozen for the online filter. The same offline fit also supplies the antenna amplitude means that seed the filter initialisation (re-centred on the reference antenna, whose elimination—not these means—fixes the gauge; Sections 2, 3), and the single thermal noise variance σ_n^2 that defines the constant i.i.d. visibility weights of Section 6: TART supplies no per-visibility weights, so σ_n^2 is estimated from a large batch of archived data, e.g. from the scatter of high-cadence differenced post-fit residuals on well-calibrated stretches (differencing adjacent frames cancels the slowly varying sky and gain contributions, leaving $2\sigma_n^2$). The correct estimator treats an offline solution track as *noisy* observations of the latent IWP and maximises the IWP-plus-observation-noise marginal likelihood (the prediction-error decomposition of the accompanying note) jointly for σ_θ^2 and the solution-noise level. A tempting shortcut—matching the empirical increment variance, $\theta_k^{(q)} - \theta_{k-1}^{(q)} \sim \mathcal{N}(0, \sigma_\theta^2 g(\Delta_k))$ with $g(\Delta)$ the relevant entry of $\mathbf{Q}_\theta(\Delta)$ —is *biased high*, because differencing noisy estimates adds twice the solution-noise variance to the increment variance; it is useful only as an initial guess for the marginal-likelihood fit. The same procedure on offline flux solutions sets the post-warm-up floor of σ_A^2 —fitted on *apparent*-flux tracks, so that the floor is high enough to follow the beam-induced elevation trend of Section 6 until the element-beam factorisation lands. Fixing these values assumes their stationarity across observations; this is reasonable for the amplitude tracks but should be checked for the phases, which can follow temperature/ionosphere—if they vary materially, a small set of regime presets (e.g. day/night) is preferable to a single global σ_ψ^2 . The fitter is a small offline utility emitting a frozen hyperparameter configuration; it never runs inside the streaming pipeline.

8 Implementation notes

The pipeline is realised as a streaming graph of separate operators—reader → sky-model → calibration → imaging → writer—so each stage has a single responsibility and can be tested in isolation:

- **Sky-model operator** computes $X_{pq,s}(t)$ from the catalogue and the per-frame geometry. TART is single-polarisation, so one scalar coherency per source/baseline suffices. The ephemeris/coordinate logic can be borrowed from TART community code (the `tart-telescope` organisation; T. Molteno’s DISKO for HEALPix imaging, and SPOTLESS—whose exact role should be confirmed before relying on it, as it appears to be an image-domain matching-pursuit deconvolver rather than a visibility-domain subtractor) [8], with `sgp4/skyfield` as the standard, well-tested route for TLE propagation and the ECEF→topocentric chain (TLE accuracy is ample here; Section 6). Those are host/NumPy implementations—we re-implement the per-frame kernels as GPU (CuPy) operators rather than calling them at runtime, but they remain the validation oracle for the coordinate conventions.
- **Calibration operator** is stateful (it carries the filter mean/covariance, the gauge, the warm-up schedule, and the fixed-lag smoother’s ring buffer of per-frame moments across frames), owns the subtraction, and emits the gain-corrected residuals (11) plus a solutions side-stream (gains, fluxes, NIS) for diagnostics and for the offline hyperparameter fit. It also owns the reference-antenna failover of Section 2: on reference-antenna flagging it resets and re-acquires with the next antenna in the configured failover order, raising a monitoring alert.
- **Imaging operator** is a new HEALPix DFT operator, structured after the existing tiled (l, m) operator, consuming residuals in the raw-visibility layout.
- **Reader** is deferred: TART data are not delivered as MSv4 Zarr, and the concrete ingest format will be fixed once sample data are in hand. The existing MSv4 reader serves only as a structural template. Whatever the format, the reader must propagate per-frame timestamps: no downstream stage may assume a constant Δ (Section 2).
- **Validation and monitoring.** The dominant source of spurious transients is not thermal noise but *imperfect subtraction*: residual sidelobes from a bright satellite or the Sun, produced by gain/flux error times a 10^5 – 10^7 Jy source, spray across the sparse-array PSF. We monitor subtraction quality and mask or down-weight pixels near bright sources, but the masking logic must discriminate between two causes of a NIS excursion. A NIS excess *localised near catalogued bright sources* indicates poor subtraction lock and is masked. A NIS excess that is *coherent across many baselines but not associated with any catalogued source* is the calibration-side signature of an unmodelled coherent emitter—i.e. a candidate transient—and must be routed to the detector as corroborating evidence, *never* used to mask the epoch: a rule that masks all epochs of elevated NIS is a transient suppressor. For correctness we validate against independent implementations: gain tracks against TART’s existing GNSS self-calibration (Scheel/Molteno), the imaging operator against DISKO on the same archived observation, and the hand-coded CuPy Jacobian against a JAX reference by finite-difference/autodiff gradient check.

The numerical backend is CuPy throughout, which also lets the calibration and imaging operators exchange GPU buffers with the existing CuPy/Holoscan tensors at zero copy. Because the calibration Jacobian is structured (antenna-bipartite gains, dense flux block), the production update is hand-coded structured linear algebra rather than a high-level autodiff framework; a JAX reference implementation of the same update is maintained for prototyping and as the gradient-check oracle noted above.

References

- [1] T. Kailath, “An innovations approach to least-squares estimation, Part I,” *IEEE Trans. Autom. Control*, 13(6):646–655, 1968.
- [2] Y. Bar-Shalom, X.-R. Li and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation*, Wiley, 2001.
- [3] S. Särkkä and A. Solin, *Applied Stochastic Differential Equations*, Cambridge University Press, 2019.
- [4] S. Salvini and S. J. Wijnholds, “Fast gain calibration in radio astronomy using alternating direction implicit methods,” *Astron. Astrophys.*, 571:A97, 2014.
- [5] M. Roth, E. Özkan and F. Gustafsson, “A Student’s t filter for heavy tailed process and measurement noise,” *IEEE ICASSP*, 2013.
- [6] K. M. Górski et al., “HEALPix: A framework for high-resolution discretization and fast analysis of data distributed on the sphere,” *Astrophys. J.*, 622:759–771, 2005.
- [7] The TART collaboration, *Transient Array Radio Telescope*, <https://github.com/tart-telescope>.
- [8] T. Molteno, DiSkO and SPOTLESS, <https://github.com/tmolteno>.